Andrew Lau

+1-415-533-7834 | andrew.lau@berkeley.edu | andrewlau.dev | github.com/andrewlau624

Education

University of California, Berkeley — Berkeley, CA
B.A. in Computer Science — *GPA: 3.8* *Expected May 2028*

Experience

Pacific | *Software Engineer Intern* — June 2026 – Present

- Cut LLM token spend 3× for coding-agent sessions with NerfGuard's optimization layer: context compression, prompt-cache-aware prefixes, and difficulty-based model rerouting.
- Designed the multi-tenant auth and admin layer: a two-tier admin API with SOC 2 controls, read-only impersonation enforced at the database and revoked on logout, API-key issuance, and MFA step-up.
- Built ingestion and ETL over entire enterprise document corpora (Microsoft 365, Exchange, Google Drive) with ACL-aware crawling, incremental sync, and GLiNER entity extraction at >95% accuracy.
- Cut prompt payloads ∼45% with hybrid retrieval, fusing BM25 lexical search with dense and sparse vectors via reciprocal rank fusion.

Apple | *Machine Learning Engineer (Contract)* — Feb. 2026 – June 2026

- Cut computer-vision validation cycle time 67% across 6 pipeline stages by building an agentic validation platform that replaced manual QA for large-scale vision training pipelines.
- Cut end-to-end validation latency 83% at 10,000+ images/day with an LLM-driven orchestration engine: a LangGraph planner compiles requests into schema-validated rules run by dependency-aware async Pub/Sub workers.
- Enabled new validation workflows without code changes via a modular model toolbox: DINOv2 and ViT quality and semantic classifiers scored on a 0–1 quality index and benchmarked against baselines to set cutoffs.

UC Berkeley EECS | *Research Software Engineer* — Aug. 2025 – Feb. 2026

- Built the therapist platform for a deployed AR/VR rehab system: an LLM pipeline turns dictated speech into step-by-step exercise prescriptions pushed to patient headsets.
- Built an LLM eval harness scoring AI session reports against therapist-labeled ground truth on step matching, outcome classification, and discomfort detection.
- Logged headset rep detection against therapist judgment to track accuracy per patient; cut deploy time ∼40%, redundant network calls ∼60%, and page loads ∼55%.

Projects

Swing Trader (Systematic Trading Platform) | *Python, pandas, Alpaca, Schwab API, systemd*

- Built a trading platform (533 tests) trading live on Schwab and paper on Alpaca from systemd timers, with broker adapters, account locking, and a wash-sale guard.
- Ran walk-forward backtests over 14,765 symbols (2,008 delisted) with block bootstrap, Newey-West t-stats, OU half-life and Hurst screens, and placebo falsification.
- Pre-registered 1,000+ experiments with kill rules; the surviving overnight-reversal leg backtests at Sharpe 2.06 / 47% CAGR / −14% max drawdown (Sharpe 1.41 at conservative costs).

agentmaxx (Coding-Agent Cost and Guardrail Layer) | *Python, Claude Code, Codex, MCP, Hooks*

- Cut Claude Code cost per task 18% (90% CI −31% to −11%) at an 18/18 hidden-test pass rate on a 6-task A/B benchmark, with 20 CLI tools, an MCP server, and session hooks.
- Replayed 339 sessions to show a fixed 86K-token prefix was 30% of billed context; cut it 63K → 17K tokens and folded repeated shell output 138KB → 1.6KB.
- Added a done-gate hook that blocks only on new test failures and a lessons hook that carried corrections into 5/5 fresh sessions vs. 0/5 without it.

Quantum RL (RL vs. Optimal Control for Qubit Gates) | *PyTorch, Gymnasium, Stable-Baselines3, QuTiP*

- Built a Gymnasium environment for transmon gate pulses and benchmarked PPO (2M steps × 5 seeds) against GRAPE, CMA-ES, and DRAG under a held-out protocol validated against QuTiP.
- Found PPO beats textbook DRAG 6–28× but trails GRAPE 2.7–200× across nominal, disturbed, and noisy settings; reported the negative result with ∼20× seed spread.

Technical Skills

Languages: Python, TypeScript/JavaScript (React, Next.js, Electron), Go, Java, SQL, C/C++, Rust
ML & AI: PyTorch, LangGraph, Claude/OpenAI SDKs, MCP, RAG (Qdrant, BM25), Stable-Baselines3, MLX, vLLM
Backend & Infra: FastAPI, PostgreSQL, Pub/Sub, Docker, GCP (Cloud Run), Argo, GitHub Actions, systemd